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Palaeomagnetism in the Coronation Geosyncline and arrangement of continents in the middle Proterozoic

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Summary. The supracrustal rocks of the Coronation Geosyncline and associated aulacogen contain upwards of 10 000 m of rocks with good stratigraphic control. In this paper new palaeomagnetic poles from four rock units are described, bringing the total from the sequence to 10. The poles are as follows: Seton Formation 02° N, 093° W, $A_{93} = 6°$, Rb/Sr isochron – 1870 Ma; Douglas Peninsula Formation 17° S, 102° W, $A_{95} = 16°$; Takiyuak Formation 13° S, 111° W, $A_{95} = 8°$; Peninsular sill 22° S, 097° W, $A_{95} = 7°$, Rb/Sr isochron – 1800 Ma. Overprinted magnetizations from the Seton and Douglas Peninsula Formations are described. The poles from the Coronation Geosyncline delineate a polar loop – the Coronation loop. Two previously observed poles from the Stark and Tochatwi do not fall on this loop and new evidence is presented which indicates that they have been affected by local rotations. Comparisons are made with results from Africa and Australia, and it is argued that there is no basis in the palaeomagnetic evidence for the proposition that the continental crust was assembled into a single unchanged pangaea during the whole of the interval – 2200 to – 1300 Ma.

Introduction

In this paper we are concerned with mid-Proterozoic events in the interval about – 2100 to – 1700 Ma. We present new palaeomagnetic results from the Coronation Geosyncline, and use these, along with previously published results, to construct a path of apparent polar wander (APW) relative to Laurentia. We then compare this path with those obtained by other workers for other continents in order to evaluate the concept of a Proterozoic supercontinent. One major problem is that of the relative rotation of small blocks. Palaeomagnetic results from such blocks yield aberrant pole positions that cannot be used to construct paths of apparent polar wandering. We have therefore sampled only from those regions that geological mapping indicates are autochthonous.

The Coronation Geosyncline (Hoffman 1973) developed in later Aphebian time (approximately – 2000 to – 1700 Ma) along the north-west border of Laurentia. Laurentia consists of the Precambrian shields of North America, Greenland and the Lewisian platform of north-west Scotland, that, apart perhaps from the Grenville Province, appears to have been a

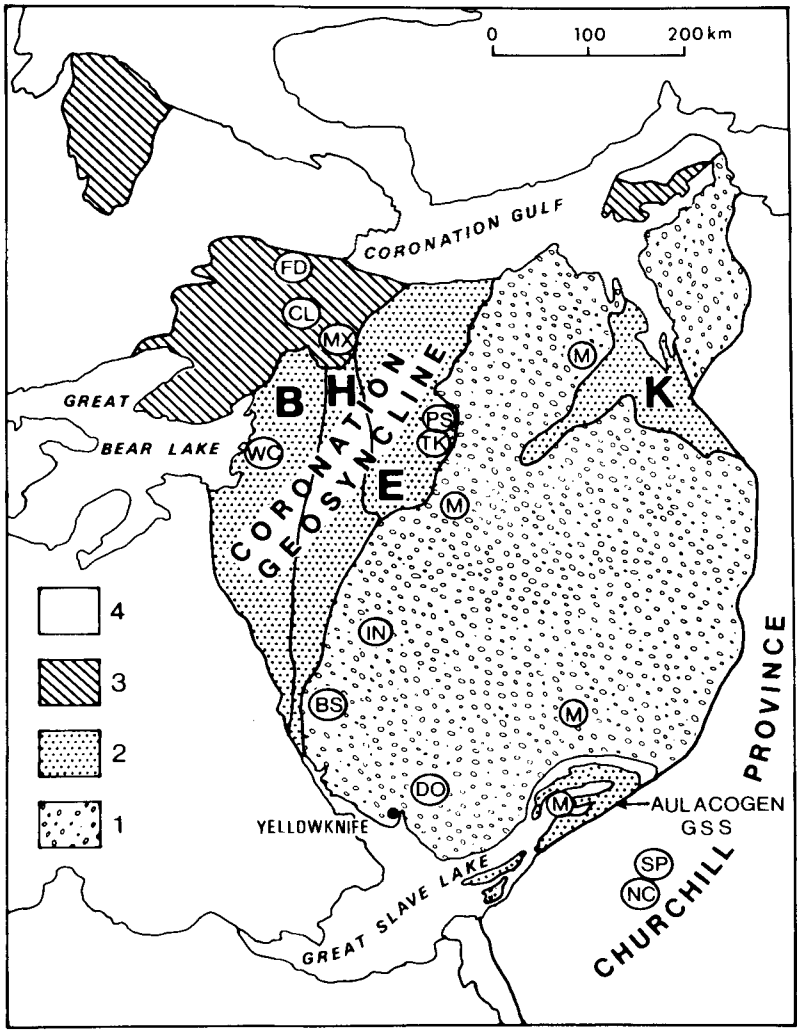


Figure 1. Regional geological map showing Coronation Geosyncline and its associated tectonic elements: B Bear Batholith, H Hepburn Metamorphic and Plutonic Complex, E Epworth Group in the Epworth basin, K Kilohigok Basin, and the Athapuscow Aulacogen that contains the Great Slave Supergroup (GSS). Ages of rocks are as follows: (1) Archaean of the Slave Structural Province, (2) Aphebian of the Coronation Geosyncline, (3) Helikian and Hadrynian cover rocks, (4) Phanerozoic rocks. Churchill Structural Province is to the east as labelled. Units that have been studied palaeomagnetically are indicated by letters in circles as follows: BS Big Spruce Complex (Irving & McGlynn 1976a), CL Coppermine lavas (Robertson 1964; Baragar & Robertson 1973), DO Dogrib dykes (McGlynn & Irving 1975), FD Franklin diabase (Fahrig, Irving & Jackson 1971; Robertson & Baragar 1972), IN Indian diabase dykes (McGlynn & Irving 1975), M Mackenzie diabase (Fahrig & Jones 1969; Irving, Park & McGlynn 1972), MX Muskox intrusion (Robertson 1969), NC Nonacho Group (McGlynn *et al.* 1974), PS Peninsular sill (herein) SP Sparrow dykes (McGlynn *et al.* 1974), WC Western Channel diabase (Irving, Donaldson & Park 1972).

more or less coherent unit of continental crust from about −2200 Ma until its breakup in the Phanerozoic (Donaldson *et al.* 1973; Stewart & Irving 1974; Irving & McGlynn 1976b). Supracrustal rocks of the Coronation Geosyncline now occur in three separate basins that were once continuous (Fig. 1); the Epworth basin, the Kilohigok Basin (Campbell & Cecile 1975) and the Athapuscow Aulacogen. The Epworth basin, the Hepburn Meta-

Table 1. Correlation chart of Coronation Geosyncline and Athapuscow Aulacogen.

Athapuscow Aulacogen	Coronation Geosyncline
Mackenzie diabase(1)(2)(3)-1250 Ma(a)	
ET-THEN GROUP(3)	
(unconformity)	
CHRISTIE BAY GROUP (GS)	Peninsular sill 1800 Ma(b)
Pearson Fm.(4)	
Portage Inlet Fm.	
Tochatwi Fm.(5)	Takiyuak Fm. (EP)(7)
Stark Fm.(6)	
(disconformity?)	
PETHEI GROUP (GS)	
Hearne Fm.	
Wildbread Fm.	
Pekanatui Fm.	
Blanchet Fm.	Cowles Lake Fm. (EP)
McLean Fm.	
Utsingi Fm.	
Taltheilei Fm.	
Douglas Peninsula Fm.(7)	
KAHOCHELLA GROUP (GS)	
Charlton Bay Fm.	
McLeod Bay Fm.(8)(9)	Recluse Fm. (EP)
Gibraltor Fm.(8)(9)	
Seton Fm.(7)-1870 Ma(c)	
SOSAN GROUP (GS)	
Akaitcho River Fm.	
Kluziai Fm.	
Duhamel Fm.	Rocknest Fm. (EP)
Hornby Channel Fm.	Odjick Fm. (EP)
(unconformity)	
UNION ISLAND GROUP	two unnamed units (EP)
(unconformity?)	
WILSON ISLAND GROUP	
(unconformity?)	
ARCHAEAN BASEMENT	ARCHAEAN BASEMENT

Compiled from Hoffman (1968, 1973). In the Athapuscow Aulacogen GS denotes the units within the Great Slave Supergroup. In the Coronation Geosyncline EP denotes the units within the Epworth Group. The Takiyuak Formation is equivalent to the Tochatwi, and possibly parts of the Stark and Portage Inlet Formations. The Cowles Lake Formation is approximately equivalent to the Pethei Group. The Recluse Formation is equivalent to the Kahochella Group, together with the Akaitcho River and Kluziai Formation of the Sosan Group. The Rocknest and Odjick Formations are equivalent to Duhamel and Hornby Channel Formations respectively. Rock units studied palaeomagnetically are in italics and numbered as follows: (1) Robertson (1969); (2) Fahrig & Jones (1969); (3) Irving *et al.* (1972); (4) McGlynn & Irving (1978); (5) Evans & Bingham (1976); (6) Bingham & Evans (1976); (7) herein; (8) McMurry *et al.* (1973); (9) Reid (1972). Preliminary results from a few samples from the McLean and Charlton Bay Formations have been given by Reid (1972). Radiometric ages are as follows: (a) K–Ar (Fahrig & Jones 1969), (b) Rb–Sr isochron (R. K. Wanless, private communication; (c) Rb–Sr Baadsgaard *et al.* (1973).

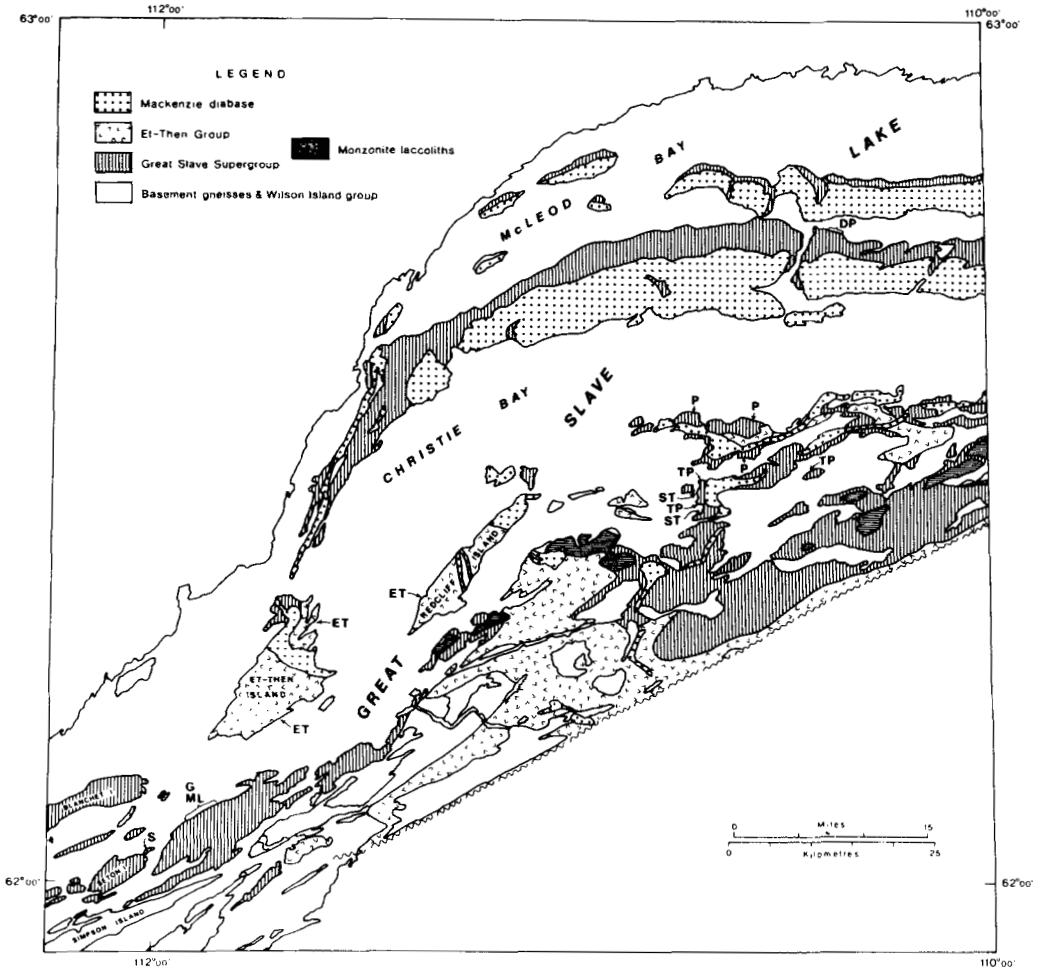


Figure 2. General sketch-map of East Arm of Great Slave Lake. Sampling areas as follows: DP Douglas Peninsular Formation (herein), ET Et-Then Group (Irving *et al.* 1972), G Gibraltar Formation (Reid 1972), ML McLeod Formation (Reid 1972), P Pearson Formation (McGlynn & Irving 1978), S·Seton Formation (herein), ST Stark Formation (Bingham & Evans 1976), TP Tochatwi Formation (Evans & Bingham 1976). Mackenzie diabase has been extensively studied in this region also (Fahrig & Jones 1969; Irving *et al.* 1972).

morphic and Plutonic Complex and the Bear Batholith comprise the Coronation Geosyncline in the strict sense (Fraser *et al.* 1972). The Kilohigok Basin and Athapuscow Aulacogen are offshoots of the main geosyncline and are included with it in a loose sense. Many thousands of metres of sedimentary rocks, spanning between 50 and 200 Ma, are present in these basins, and they offer an excellent opportunity of obtaining a stratigraphic sequence of palaeomagnetic observations. In Table 1 a correlation chart is given, and previous palaeomagnetic studies indicated. The new results are from the Seton and Douglas Peninsula Formations of the Athapuscow Aulacogen, and from the Takiyuak Formation and a thick gabbro sill in the Epworth basin, so that observations from different parts of the geosyncline can now be compared. The sampling localities are shown in Figs 2 and 3.

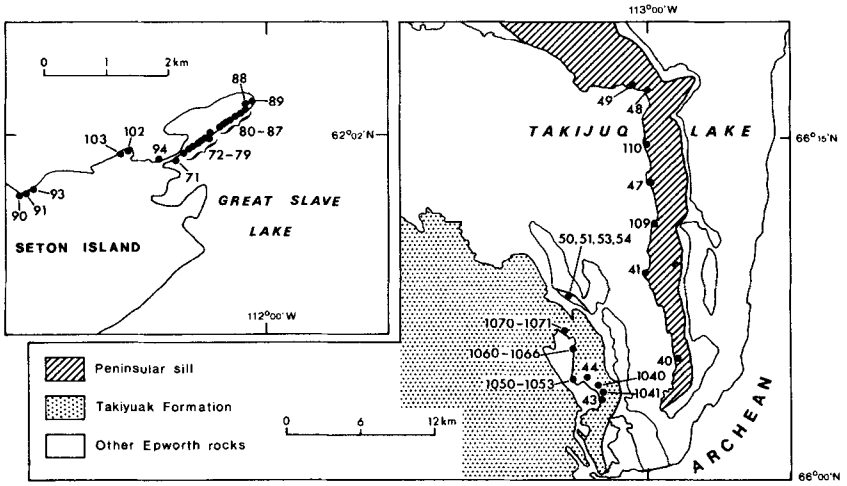


Figure 3. Sketch-maps of sampling locations for the Seton Formation (left), and the Peninsular sill and Takiyuak Formation (right, geology from Fraser 1974).

Procedures

Usually five cores were obtained at each collecting site, and from most cores two cylindrical specimens were cut. The magnetization was measured by Schonstedt DSM magnetometers. Alternating field (af) and thermal demagnetization was carried out using the apparatus described by Roy, Sanders & Reynolds (1971), Roy, Reynolds & Sanders (1973), and a Schonstedt thermal demagnetizer. Most demagnetizations were carried out in ambient fields of less than 2 nT using the field cancellation system of Roy, Robertson & Keeping (1969). Analysis was made using Fisher's statistics (Fisher 1953). In sedimentary rocks, cores were taken from a recognizable bed, usually no more than 0.5 m thick, and to obtain mean directions unit weight was given to individual specimens, because specimens were from different levels within the bed and probably recorded the field at different times. In igneous rocks, samples were spread over an available outcrop, say over 10 m, and unit weight was given to each core, results from specimens being averaged because during cooling there is likely to be negligible time difference between the record of adjacent specimens. To obtain average directions of rock bodies unit weight was given to sites. When reversals are present the calculations were made irrespective of sign.

Seton Formation

The Seton Formation belongs to the pre-flysch phase of the Kahochella Group. It consists predominantly of volcanogenic rocks that occur at various stratigraphic levels within the sedimentary formations of the Kahochella and Sosan Groups (Hoffman 1968, 1973 and Hoffman *et al.* 1977). Baadsgaard, Morton & Olade (1973) have reported a Rb–Sr isochron age of -1870 ± 10 Ma ($\lambda = 1.39 \times 10^{-11}\,\mathrm{yr}^{-1}$) from the type section on Seton Island.

At the type section, vitric tuffs and red tuffaceous siltstones were collected from the lower third of the formation, and massive andesite, red siltstones and basic tuffs from near the top of the sequence (Fig. 2). The thickness of the Seton Formation at the type section is about 1290 m, and Hoffman *et al.* (1977) suggest that this part of the Seton is, in part, equivalent in age to the Kluziai and Akaitcho River Formations of the Sosan Group. Also, according to Hoffman *et al.* (1977), the section is essentially autochthonous. The strike of

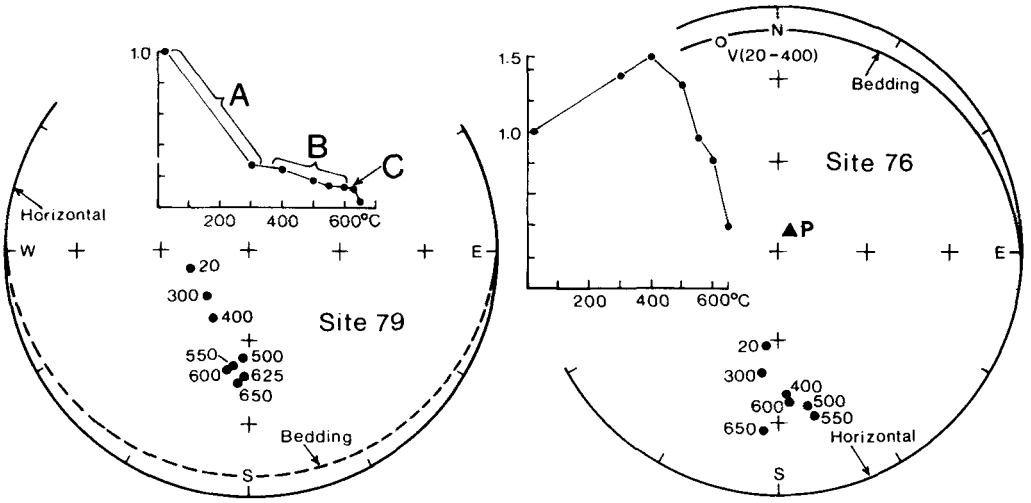


Figure 4. Thermal demagnetization of Seton Formation. The decay curve and corresponding directions labelled with the temperatures (°C) are shown. Downward (upward) inclinations are denoted by dots (circles). P is the present field direction. The trace of the bedding plane is shown, dashed when on the upper hemisphere. At site 79 the parts of the decay curve corresponding to the A, B and C magnetization are shown. At site 76 V(20–400) is the direction of the magnetization removed between 20 and 400°C; note that it is approximately reversed from the high-temperature magnetization.

bedding is usually 70–90°, with gentle dips of 10–20° both for the north and south. Polished sections of the sedimentary rocks show that hematite is the only opaque mineral present. It occurs as a fine dusting on sedimentary grains, as ochre in the matrix, and as detrital grains 10–15 μm in diameter. The flows consist of sodic plagioclase and chlorite, along with minor amounts of iron oxides, sphene and carbonate. The assemblage is compatible with, but not diagnostic of, the prehnite-pumpellyite facies of metamorphism, similar to that in lavas of the Pearson Formation at the top of the Great Slave Supergroup (McGlynn & Irving 1978), and could have been formed either deuterically just after crystallization, or during later burial.

In the left side of Fig. 4 a typical thermal demagnetization result is shown. The initial magnetization (nrm) is steeply inclined downwards, and the direction migrates to the south, reaching a good end-point between 550 and 650°C. The magnetization removed below 400°C is referred to as *A* magnetization. The directions of the *A* magnetizations, obtained by vector subtraction, are plotted in Fig. 5, with the breakdown by sites in Table 2. The directions are close to the present field, and the summary statistics in Table 3 show that their mean does not differ significantly from it. The *A* magnetization could be a viscous magnetization imposed by the present field (030, +82). The magnetizations removed between 400 and 600°C are directed to the south-east, and differ significantly from the present field (Fig. 5). This is referred to as the *B* magnetization, which is interpreted as an overprint acquired during uplift following deep burial (see below). The *B* magnetization is parallel to the overprint found by McMurry, Reid & Evans (1973) in the Gibraltar and McLeod Bay Formations that overlie the Seton (Table 1). The magnetization remaining at 600°C or above (referred to as the *C* magnetization) usually is well-grouped and directed toward the south-east with intermediate inclinations (Fig. 5). After cleaning, all specimens at sites 79 and 82, which have typical *C* magnetization directions, were cut in two and re-measured. There was excellent agreement, generally better than 5°, showing that the specimens were homogeneously magnetized. The directions of the *C* magnetization at sites

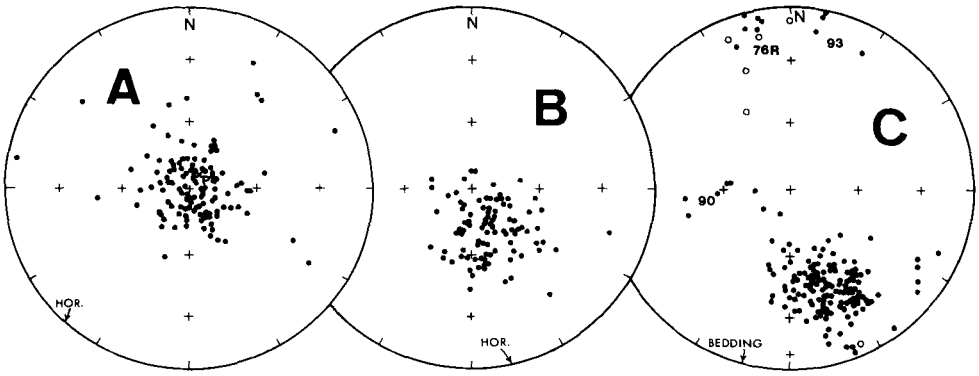


Figure 5. Magnetization directions in the Seton Formation. Directions observed in all specimens are plotted. The magnetization removed in the intervals 20 to 400°C (A), and 400 to 600°C (B) are shown at the left and centre. On the right the directions of the C magnetizations obtained after magnetic cleaning in temperature of about 600°C (Table 2) are shown. The directions at sites 76R, 90 and 93 are labelled separately. P is the direction of the Earth's present field.

Table 2. Results from the Seton Formation.

Site	N(S)	M	D,I	k	α_{95}	Ref.	Treatment °C
71A	9(5)	9	274,+83	33	9	h	V(20-400)
71B	3(2)	-	148,+65	96	13	h	V(400-600)
71C	((5)	-	147,+45	27	10	b	cleaned 600
72A	4(4)	15	128,+75	32	17	h	V(20-400)
72B	6(4)	-	113,+56	29	13	h	cleaned 625
73A	4(3)	60	039,+81	26	18	h	V(20-400)
73B	8(5)	-	145,+64	38	9	h	cleaned 600
74B	9(5)	22	129,+80	47	8	h	cleaned 600
75A	5(4)	27	269,+82	13	22	h	V(20-400)
75C	8(5)	-	156,+64	65	7	h	cleaned 650
76C(r)	8(5)	-	344,-21	26	11	b	V(20-400)
76C(n)	9(5)	-	162,+46	158	4	b	cleaned 400
77A	10(5)	42	141,+83	58	6	h	V(20-400)
77B	5(3)	-	140,+59	55	10	h	V(400-600)
77C	10(5)	-	150,+42	126	4	b	cleaned 600
78A	5(3)	27	358,+85	42	12	h	V(20-400)
78C	10(5)	-	144,+40	89	5	b	cleaned 600
79A	8(4)	39	289,+80	36	9	h	V(20-400)
79B	8(4)	-	204,+72	21	12	h	V(400-600)
79C	8(4)	-	182,+65	84	6	b	cleaned 600
80A	10(5)	33	321,+84	11	16	h	V(20-400)
80B	9(5)	-	162,+60	28	10	h	V(400-600)
80C	10(5)	-	171,+54	54	7	b	cleaned 625
81A	10(5)	14	356,+69	4	28	h	V(20-400)
81B	8(5)	-	197,+64	15	15	h	V(400-600)
81C	10(5)	-	141,+46	49	7	b	cleaned 600
82A	4(4)	79	150,+71	156	7	h	V(20-400)
82C	5(5)	-	155,+57	216	5	b	cleaned 650

Table 2 – *continued*

Site	N(S)	M	D,I	k	α_{95}	Ref.	Treatment °C
83C	9(5)	21	163,+47	33	9	b	cleaned 625
84A	4(4)	41	114,+65	13	27	h	V(20-400)
84B	7(5)	-	180,+59	34	11	h	cleaned 625
85C	8(5)	24	157,+48	59	7	b	cleaned 600
86A	4(3)	27	355,+81	21	21	h	V(20-400)
86B	4(3)	-	165,+54	31	17	h	V(400-600)
86C	8(5)	-	152,+37	30	10	b	cleaned 600
88C	10(5)	77	159,+59	43	12	b	cleaned 530
89C	5(4)	695	157,+47	13	25	b	cleaned 500
90A	8(5)	44	40,+61	9	20	h	V(20-400)
90B	7(4)	-	156,+69	14	17	h	V(400-600)
90C	9(5)	-	277,+60	22	11	b	cleaned 600
91A	7(5)	79	173,+83	28	12	h	V(20-400)
91B	7(5)	-	163,+80	27	12	h	cleaned 600
93A	8(4)	29	024,+83	26	11	h	V(20-400)
93B	7(4)	-	132,+82	39	10	h	V(400-600)
93C	6(4)	-	007,-04	20	15	b	cleaned 600
94A	4(4)	16	176,+85	31	16	h	V(20-400)
94C	4(4)	-	162,+58	126	8	b	cleaned 400
102A	11(6)	73	125,+76	8	25	h	V(20-400)
102B	11(6)	-	162,+60	65	8	h	V(400-600)
102C	11(6)	-	160,+40	16	12	b	cleaned 600
103C	6(4)	8	145,+40	17	17	b	cleaned 625

N is the number of specimens and S the number oriented cores. M is the intensity of natural remanent magnetization $\times 10^{-3}$Am. D and I are the declination and inclination of the magnetization, k is Fisher's precision and α_{95}° the error in the direction ($P = 0.05$). The site numbers are labelled A, B or C according to the magnetizations represented. The directions are referred to the horizontal (h) or bedding (b) plane as noted in the column headed ref. In the right column the procedure for obtaining the magnetization is given. V denotes magnetization removed in the temperature interval indicated and obtained by vector subtraction. Other directions obtained by thermal cleaning at the temperature indicated.

Table 3. Summary of results.

		B	N	D,I	k	α_{95}	Pole
Seton Formation	A poles (h)	18	115	-	11	11	100W,64N
	B directions (h)	14	99	158,+68	36	7	097W,24N
	B poles (h)	14	99	-	17	10	097W,25N
	C directions (b)	19	154	159,+46	24	7	093W,01N
	C poles (b)	19	154	-	29	6	093W,02N
Douglas Peninsula Formation	LT directions (h)	10	54	233,+41	125	4	158W,06N
	LT poles (h)	10	54	-	119	4	158W,06N
	HT directions (b)	6	43	172,+18	14	19	102W,18S
	HT poles (b)	6	43	-	19	16	102W,17S
Takiyuak Formation	directions (b)	17	111	178,+19	15	10	111W,14S
	poles (b)	17	111	-	20	8	111W,13S
Peninsular sill	directions (b)	7	32	345,-02	63	8	097W,22S
	poles (b)	7	32	-	73	7	097W,22S

B is the number of sites, N the number of specimens (sedimentary rocks) or samples (two specimens each in igneous rocks). Other symbols as in Table 2.

76, 90 and 93 do not fall in the main distribution. At site 90 the directions are westerly and remain there up to 625°C. Three specimens from site 90 were split in half and re-measured. One pair remained little changed in direction, but one of each of the second and third pairs had a southerly direction comparable to the typical *C* magnetization. Hence the magnetization at site 90 consists of components with different directions that have not been resolved thermally. At site 93 northerly directions were observed. They are stable up to 665°C and splitting showed them to be homogeneously magnetized. At site 76 the magnetizations removed between 20 and 400°C were not along the present field, but had directions similar to those at site 93 (site 76R, Table 2, Fig. 5). The decay curve and corresponding direction are shown on the right side of Fig. 4. Thus at site 76 there are two antiparallel magnetizations, a northerly direction with blocking temperatures between 20 and 400°C, and a southerly direction with blocking temperatures between 400 and 650°C. Apparently both 76R and 93 are genuine reversed magnetizations.

The mean directions of the *A*, *B* and *C* magnetizations are given in Table 3. The dip contrasts are insufficient to provide a fold test; Fisher's precision k increases from 21 to 24 after correction for folding, which is not a significant change. The direction in the andesite flow within the Seton (site 89, Table 2) agrees well with those in the sedimentary rocks. We interpret the *C* magnetization to be the original magnetization acquired between the time of deposition and lithification.

Douglas Peninsula Formation

The Douglas Peninsula Formation, the lowest formation of the Pethei Group, occurs throughout the aulacogen and varies in thickness from 16–33 m (Hoffman 1968). The type section was sampled. It is 25 m thick and it is autochthonous (Hoffman *et al.* 1977). The rock is a thin-bedded, fissile, red marlstone, that is composed of small lenticles of pink or brown limestone in dark reddish-brown mudstone. The dominant opaque mineral is haematite, which occurs as a dusting on sedimentary grains, and as very fine blebs on sand grain boundaries. There are detrital grains of haematite, up to 50 μm in diameter. Haematite, leucoxene and rare remnants of ilmenite, occur in the matrix as blocky grains that range in diameter from 10–50 μm. Because of its fissile nature it was very difficult to sample, and at some levels unbroken core could not be obtained. The 12 sampling sites are therefore distributed unevenly through the section. Sites 57 and 61 are from a prominent mudstone bed.

The initial (nrm) directions were very well-grouped in the south-west quadrant close to the direction observed in Mackenzie diabase. The diabase occurs as huge sheets intruding the Pethei Group in the sampling area. Behaviour during thermal demangetization is shown in Fig. 6. At site 52 the directions remain in the south-west quadrant, essentially unchanged up to 625°C. This behaviour was found at five sites, although commonly the magnetization was lost at temperatures above 500°C. At the other sites the directions migrated during demagnetization, either to the south or to the north. Apparently three magnetizations are present, one with blocking temperatures predominantly below 400°C directed toward the south-west, and two with blocking temperatures exceeding 600°C, one directed to the north and the other to the south (Fig. 7). The low temperature magnetization constitutes between 90 and 100 per cent of the nrm. We believe that this dominant magnetization is a secondary overprint acquired during cooling following intrusion of the nearby Mackenzie diabase. The high temperature magnetization is present at six sites, and never constitutes more than 10 per cent of the nrm. The high temperature magnetization directions are not exactly reversed from one another, probably owing to incomplete separation of truly antiparallel

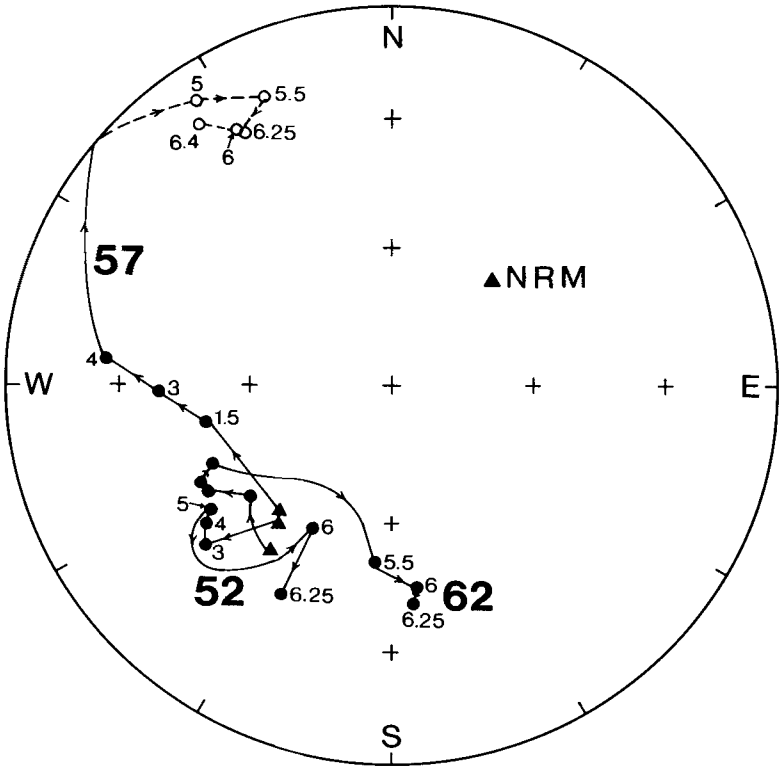


Figure 6. Direction changes observed during thermal demagnetization of specimens from sites 52, 57 and 62 of the Douglas Peninsula Formation. Each point is labelled with the temperature in °C (× 10^{-2}).

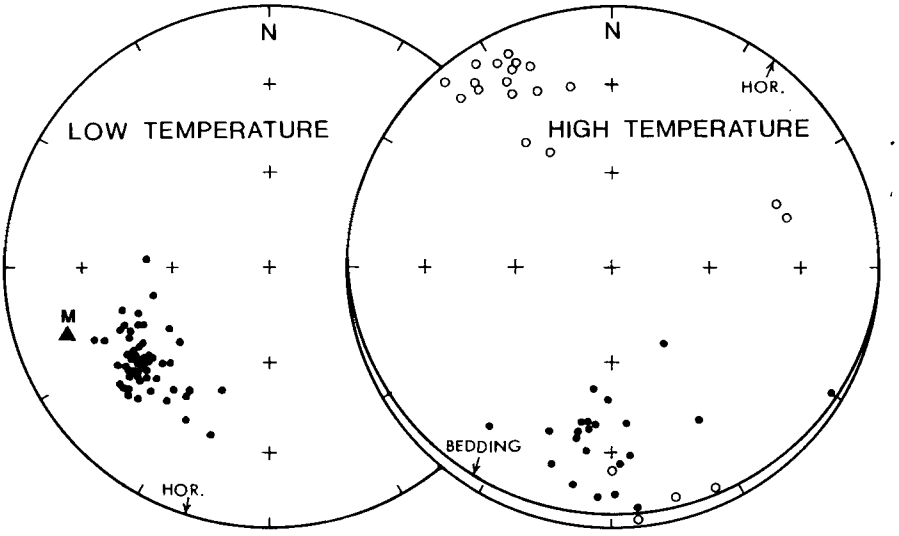


Figure 7. Directions observed in specimens from the Douglas Peninsula Formation after thermal cleaning. On the left the directions after cleaning at 200 to 400°C, and on the right after cleaning at 600°C or above, are shown. M is the mean direction observed from Mackenzie diabase in the sampling area (Irving *et al.* 1972).

Table 4. Results from the Douglas Peninsula Formation.

Site	N(S)	M	D,I	k	α_{95}°	h	T°C
48L	4(3)	13	228,+39	144	8	6.2	400
49L	5(5)	10	232,+44	49	11	7.4	300
50L	7(5)	8	229,+36	85	7	8.0	200
51L	5(5)	13	240,+46	126	7	8.9	300
51H	5(5)	-	194,+30	17	19	-	625
52L	5(5)	8	231,+37	223	5	9.8	300
53L	5(5)	7	246,+36	122	7	10.2	300
54L	5(5)	7	228,+37	79	9	10.3	300
54H	5(5)	-	187,+08	31	14	-	640
55L	7(5)	6	230,+37	41	10	10.8	300
55H	7(5)	-	160,-03	9	22	-	600
56L	6(6)	11	242,+47	25	14	13.2	400
57H	9(6)	8	341,-24	4	29	18.3	625
61H	9(5)	13	330,-12	29	10	18.3	625
62L	5(5)	10	225,+48	46	11	19.0	400
62H	9(5)	-	182,+34	35	9	-	650

In the first column, L and H refer to the low- and high-temperature magnetizations. The L directions are given with respect to present horizontal. The H directions are corrected for a southerly dip of 7°. h is the distance in metres below the top of the formation, and T is the cleaning temperature in °C. Other symbols as in Table 2.

components. Errors from this source are minimized by taking the mean irrespective of sign (Cox & Doell 1960). Reversed magnetization occurs only in the prominant mudstone bed 18.3 m from the top of the formation, with normal magnetization above and below.

Takiyuak Formation

The Epworth Group spans three tectonic zones of the Coronation Geosyncline, namely the western zone of metamorphic rocks, a central zone of thrust sheets, and an eastern autochthonous zone. The Takiyuak Formation occurs only in the autochthonous zone, and is the youngest unit of the Epworth Group (Table 1). It is the molasse phase of the geosyncline, and comprises immature terrigenous sediments deposited in an alluvial environment (Hoffman 1973). The total thickness is about 600 m, and in the sampling area the beds have a regional dip of 10° to the south-west. The basal units consist of red mudstones and siltstones, which grade upward into mudstones and thickly bedded lithic sandstones. Lithic clasts include clastic and carbonate sediments, silicic volcanic rocks and plutonic and metamorphic rocks. Haematite is the most abundant oxide, occurring as dust on silicate sedimentary grains, as fine blades (5 μm or less) distributed around sedimentary grains, and as rounded detrital grains up to 300 μm in diameter composed of crystalline specular haematite or amorphous or earthy haematite. Haematite also occurs in the lithic fragments, in which the grains have lamellae of haematite and leucoxene with octahedral texture, which probably results from the alteration of grains of magnetite with ilmenite lamellae, during sedimentation.

The nrm directions are always grouped near the present field. Examples of behaviour during thermal demagnetization are shown in Fig. 8. At site 107 the direction is steeply

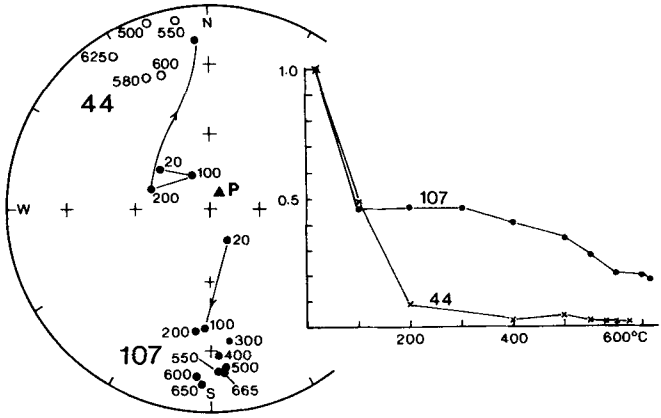


Figure 8. Thermal decay curves and corresponding direction changes in specimens from sites 44 and 107 of the Takiyuak Formation. The points are labelled with the temperature in °C. Open circles denote upward inclinations.

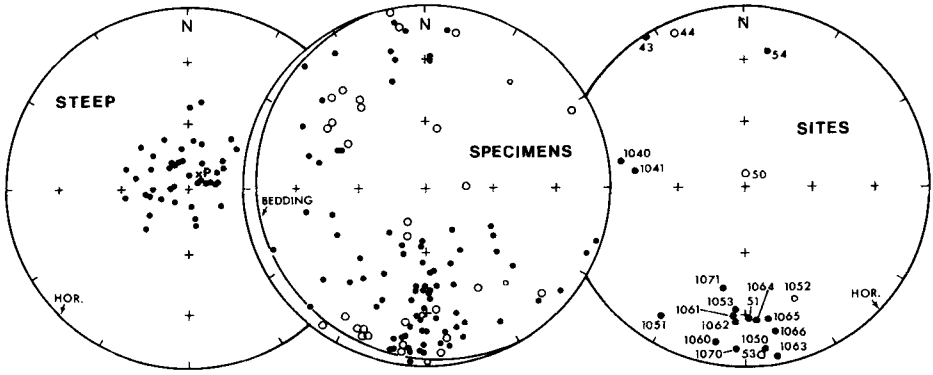


Figure 9. Directions observed in red sandstones from the Takiyuak Formation after thermal cleaning at 600 to 650°C. On the left, directions that remain around the Earth's field (P) during thermal demagnetization, are shown. In the middle, all other specimen directions are plotted, and on the right, mean directions at each collecting site.

inclined downwards, moving quickly to the south with a reasonable end-point between 500 and 665°C. Such well-defined behaviour, however, is rare. For example, at site 44 the decay of intensity is very rapid, and the magnetization at 600°C constitutes only 2 per cent of the nrm. The directions move toward the north with shallow negative inclination, but there is more scatter, the directions undergoing random fluctuations with amplitudes of about 30°; a consistent but weak signal is present. Generally the intensity after magnetic cleaning was between 2 and 0.2×10^{-3} A/m. The overall precision and the precision within sites is generally poor (Fig. 9, centre). There are also numerous specimens whose directions remain clustered around the present field during thermal demagnetization up to 600°C, and these could be due to recent weathering (Fig. 9, left side). The mean site directions (excluding the directions close to the present field) show a predominantly southerly grouping. There are also three sites with directions to the north, and three sites with intermediate directions (Fig. 9, right side). Splitting tests, to determine whether these three intermediate directions arose from mixing of unresolved components, were not successful, because the intensities

Table 5. Results from the Takiyuak Formation.

Site	N(S)	M	D,I	k	α_{95}	T°C
43	4(4)	54	326,+00	6	43	625
44	5(5)	62	335,-04	4	41	600
50	7(5)	2	000,-84	5	29	600
51	4(3)	53	178,+29	5	47	600
53	9(5)	49	196,-08	16	13	600
54	4(3)	46	008,+25	12	28	600
1040	5(3)	40	281,+31	20	17	650
1041	4(2)	38	278,+39	5	47	625
1050	6(3)	24	173,+11	4	40	650
1051	4(2)	22	213,+16	9	32	600
1052	4(3)	28	156,+33	5	49	600
1053	3(2)	21	185,+32	11	40	600
1060	8(5)	26	190,+14	3	35	600
1061	5(3)	39	195,+29	42	12	600
1062	5(3)	31	184,+26	5	36	600
1063	3(3)	29	169,+05	4	68	625
1064	4(3)	35	175,+27	3	67	600
1065	6(3)	39	170,+26	5	33	600
1066	3(2)	37	168,+20	7	50	600
1070	10(5)	28	184,+11	80	5	650
1071	8(4)	32	192,+42	11	17	600

T is the cleaning temperature (°C). Other symbols as in Table 2. All directions are corrected for regional tilt of 10° in a direction 242°.

of the divided specimens after cleaning were below detection levels. The breakdown by sites and the overall average direction are given in Tables 3 and 5. Because of the high blocking temperatures and the presence of reversals, we interpret this as being the original magnetization acquired during deposition and lithification.

Peninsular sill

The Peninsular sill cuts the Epworth Group, and appears to have been intruded before folding (Fig. 3). The sill dips at about 10° to the west and is over 40 m thick. It is a fine- to medium-grained gabbro, comprising pyroxene and basic plagioclase lathes, minor quartz, less than 1 per cent biotite and about 1 per cent iron oxides. Plagioclase is slightly altered to white mica and zoisite, and pyroxene grains are partly altered to brown hornblende or green actinolite and chlorite. Coarser grained phases at the top of the sill contain more biotite, minor quartz, myrmekitic intergrowths and very minor amounts of potash feldspar, suggesting that the sill is somewhat differentiated. A Rb–Sr isochron of – 1800 Ma has been obtained (R. K. Wanless, private communication, $\lambda = 1.39 \times 10^{-11}\,\mathrm{yr}^{-1}$).

Ilmenite is the most abundant oxide. In the coarser fractions it occurs as discrete grains up to 2000 μm in diameter. Magnetite occurs as rims or grains around the ilmenite grains. There is some alteration of ilmenite to leucoxene and magnetite to haematite. Haematite also occurs as discrete grains.

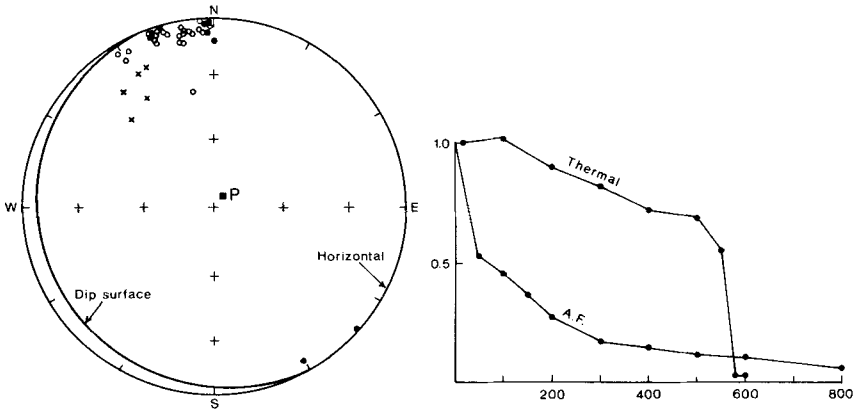


Figure 10. Peninsula diabase. The mean sample directions (two specimen directions per core sample) with respect to horizontal observed after magnetic cleaning are given on the left. The dip surface of the sill is indicated. Open circles denote negative inclinations, solid circles positive inclinations and crosses downward directions from baked contact sediments at site 110. P is the Earth's present field. Thermal and af demagnetization curves at site 41 are given on the right. The horizontal scale gives the peak alternating magnetic field in units of 0.1 mT (1 mT = 10 oersted) and the temperature in °C. The initial directions migrate to the north-north-west with low inclination and remain there until 100 mT and 550°C respectively.

Demagnetization curves are given in Fig. 10. With one exception (site 110) the behaviour is simple. After cleaning in fields of about 20 to 30 mT or heating to 550°C the directions are very well grouped (Table 6). At site 110, which is at the chilled margin, two antiparallel magnetizations occur. Sediments sampled within 1 m of the margin are weakly magnetized after demagnetization (0.8×10^{-3} A/m), and they give directions comparable to those of the sill, but with somewhat steeper inclinations. This phenomenon of divergent directions in baked contact rocks has been well documented by Park (1974). Its cause is not certain; it is possible that during cooling the Earth's field at the contact was disturbed by the magnetization of the sill itself; it is also possible that the iron minerals in sill and sediment were being chemically altered during cooling and that this alteration was arrested at similar but not identical times.

Table 6. Results from the Peninsular sill.

Site	S	M	D,I	k	α_{95}°	Treatment
40	5	378	350,-06	24	16	30mT
41	5	222	351,-04	381	4	30,40mT
47	5	145	356,+02	313	4	30mT
48	5	313	348,+07	50	11	20mT
49	4	109	341,-02	64	12	20mT, 550°C
109	4	214	343,-05	73	1	30,40mT
110S	5	9	332,+26	72	9	30mT, 400°C
110R	2	18	336,-02	194	-	20mT, 550°C
110N	2	23	141,+05	32	-	100mT, 550°C

All directions are corrected for a tilt of 10° in a direction 242°. *S* is the number of oriented cores, two specimens per core. The treatment in alternating fields and by heating is indicated in the right column. Other symbols as in Table 2.

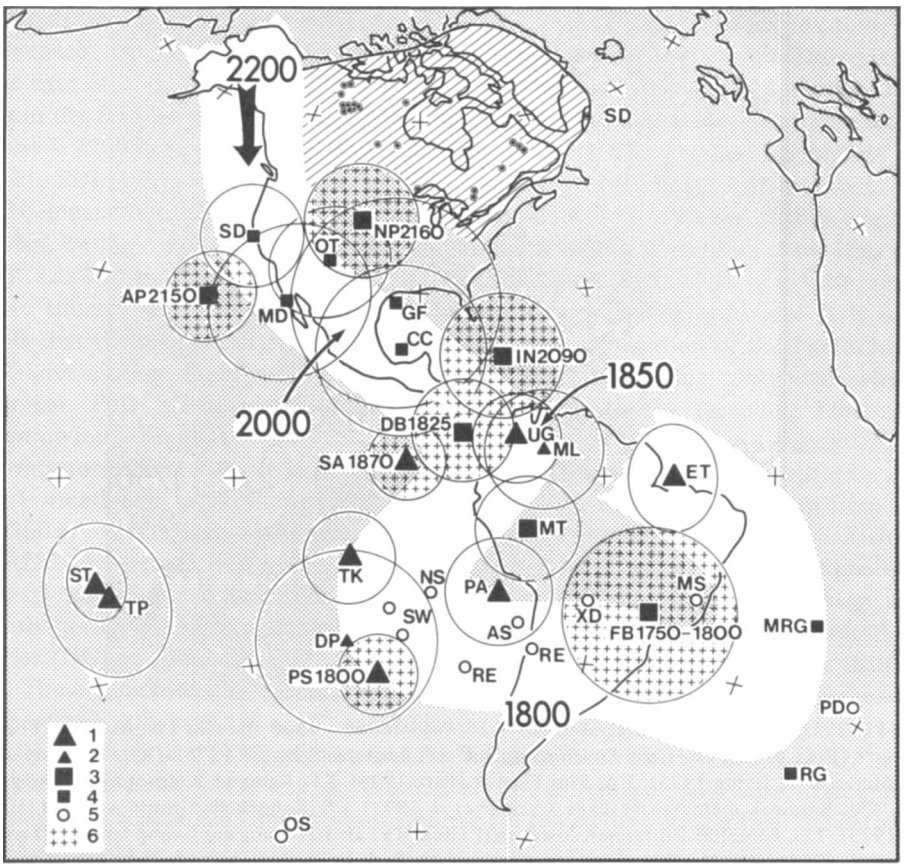


Figure 11. Pole path for the interval about – 2200 to – 1750 Ma. The pole path has been compiled mainly from Irving & McGlynn (1976b) and Roy & Lapointe (1976) and the results herein. The symbols for poles are as follows: (1) well-defined poles from the Great Slave Supergroup, (2) poles less well defined from the Great Slave Supergroup, (3) ((4)) well-defined (less well-defined) poles from the other areas of the Canadian Shield, (5) overprints, (6) poles that are probably primary from rocks with good age control. AP, Abitibi dikes primary magnetization (Larochelle 1966; Irving & Naldrett 1977) – 2147 Ma. AS, Abitibi dikes secondary magnetization. CC, Castignon Complex (Park 1977), – 1875 Ma, Dressler (1975). DB, Dubawnt Group (Park, Irving & Donaldson 1973) – 1830 Ma. DP, Douglas Peninsular presumably primary magnetization (herein). ET, Et-Then Group (Irving *et al.* 1972). FB, Flin Flon Greenstone belt *B*, magnetization primary in Boundary Intrusions (Park 1975) – 1800 Ma. GF, Gunflint Formation (Symons 1966) *c.* – 2000 Ma. IN, Indin dikes (McGlynn & Irving 1975) – 2093 Ma. MS, Menihek Formation Labrador Trough secondary magnetization (Park 1977) *c.* – 1800 Ma. ML, McLeod Formation (Reid 1972). MRG, Richmond Gulf and Manitounuk volcanic rocks (Schwarz 1976, 1977). MD, Molson dikes (Ermanovics & Fahrig 1975) *c.* – 2000 Ma. MT, Martin Formation (Evans & Bingham 1973) *c.* – 1850 Ma. NP, Nipissing diabase presumably primary magnetization (Roy & Lapointe 1976) – 2160 Ma. NS, Nipissing diabase secondary magnetization (Roy & Lapointe 1976). OT, Otish gabbro (Fahrig & Chown 1973) *c.* – 2000 Ma. PA, Pearson Formation A magnetization (McGlynn & Irving 1978) *c.* – 1800 Ma. PD, Pearson Formation D secondary magnetization (McGlynn & Irving 1978). PS, Peninsula sill (herein) – 1800 Ma (Wanless, private communication). RE, Retty peridotite Labrador Trough, two results, probably secondary (Park 1977). RG, Richmond Gulf red beds and andesites (Schwarz 1976, 1977). SA, Seton Formation A magnetization (herein) – 1800 Ma. SD, Scourie dikes of north-west Scotland corrected for closure of Atlantic (Beckman 1976), age of intrusion – 2200 to – 1900 Ma, possible age of magnetization – 1800 Ma. See text. ST, Stark Formation (Bingham & Evans 1976). SW, Seward Formation A magnetization, secondary (Park 1977). TK, Takiyuak Formation (herein). TP, Tochatwi Formation (Evans & Bingham 1976). UG, Gibraltar Formation upper part (Reid 1972). XD, 'X' Dykes (McGlynn & Irving 1975).

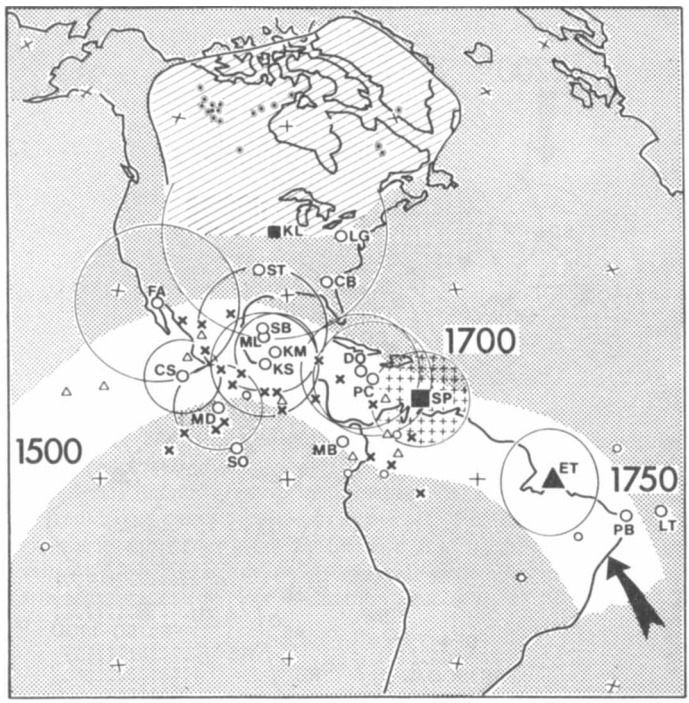


Figure 12. Polar path for the interval about −1750 to about −1500 Ma. CB, Charlton Bay Formation overprint? (Reid 1972). CS, Cape Smith overprint? (Fujiwara & Schwarz 1975). DO, Dogrib dykes overprint (McGlynn & Irving 1975). FA, Flin Flon *A* (Park 1975). KL, Kaminak lamprophyre (Christie *et al.* 1975). KM, Kaminak metadiabase dikes (Christie *et al.* 1975). KS, Kahochella Group overprint (McMurry *et al.* 1973). LG, Gibraltar Formation lower part (Reid 1972). LT, Dubawnt Group lapilli tuff overprint (Park *et al.* 1973). MB, Menihek Formation overprint? (Park 1977). MD, Melville-Daly Bay metamorphic rocks (Park 1973). ML, McLean Formation overprint? (Reid 1972). PB, Pearson Formation B (McGlynn & Irving 1978). PC, Pearson Formation C (McGlynn & Irving 1978). SB, Seton Formation overprint? (Reid 1972). SO, Seward Formation overprint (Park 1977). SP, Sparrow dikes 1700 Ma (McGlynn *et al.* 1974). Symbols for poles as in Fig. 11. In addition, Nonacho Group secondary magnetizations are denoted by open triangles (McGlynn *et al.* 1974), the Big Spruce overprints by open circles without error circles (Irving & McGlynn 1976a) and the Nagsugsugtoquidian dikes secondary magnetizations by diagonal crosses (Morgan 1976).

Apparent polar wander and the problem of local rotations

The results are summarized in Table 3. The poles are plotted in Figs 11 and 12, along with all others from Laurentia obtained for the time interval from about −2100 to −1600 Ma. In the review of previous results a first-order filter was applied. Results based on fewer than 10 samples or with an error greater than 20° (minimum reliability criteria) are not used. They constitute about 5 per cent of the data. The proposed average polar path is shown as a broad swath roughly 20° in width made necessary by the polar errors. The results from the Coronation Geosyncline are indicated by triangles in Fig. 11 and their stratigraphic order is given in Table 1. They are preceded by a group of poles (AP, NP, OT, MD and SD) dated at about −2100 Ma, derived from tholeiitic dykes, sills and plutons that are widespread in Laurentia at that time. These have been referred to as the post-Huronian igneous events (Roy & Lapointe 1976). Two of the poles (Abitibi dykes AP, and the Nipissing diabase NP) have reliable associated Rb–Sr isochrons. They are also based on fully executed contact tests in which the magnetization directions in country rock near to the intrusions are parallel

to those in the intrusions themselves, but different from the directions in the country rock at a distance; hence the magnetization was acquired at the time of intrusion. The constraints on the ages of poles MD, OT and SD are less good. Especially uncertain is the age of pole SD from the Scourie dykes. Radiometric evidence indicates that these dykes were intruded between −2200 and −1900 Ma (Beckman 1976, table 1). The pole for the Scourie dykes (Fig. 11, SD) is in good agreement with poles of the post-Huronian igneous events which are of comparable age, and the magnetization could date from the time of their intrusions. Beckman, on the other hand, argues that the magnetization was acquired at the time of the nearby Laxfordian deformation (−1850 Ma). The dykes themselves are 'little altered', and are situated away from the zone of intense Laxfordian deformation. It is equally possible that the magnetization was acquired not at the culmination of the Laxfordian but much later during final cooling following the regional heating associated with the Laxfordian. Pole SD may therefore rightfully belong to the ascending limb of the Coronation loop (Fig. 12).

Moving forward in time it is noted that the pole (SA) for the *A* magnetization of the Seton Formation is in excellent agreement with that for the Dubawnt Group from Baker Lake 700 km to the east. These two rock units have comparable Rb–Sr ages (−1870 ± 10 and −1830 ± 5 respectively). The Dubawnt Group rests unconformably on Aphebian or Archaean basement and is certainly autochthonous. This suggests that the Seton is also autochthonous, as the geological evidence indicates (Hoffman *et al.* 1977). The poles for the Gibraltar (UG) and McLeod (ML) Formations fall nearby, but are displaced a little eastwards. These formations have been sampled on Keith Island, and the polar difference could be interpreted as caused by a 15° anticlockwise rotation of Keith with respect to Seton Island. Hoffman *et al.* (1977) suggest that the Kahochella on Keith Island is allochthonous, being part of a north-facing nappe. Alternatively the difference could arise from apparent polar wander, because the Gibraltar and McLeod are somewhat younger than the Seton that we have sampled.

The poles from the younger rocks of the Athapuscow Aulacogen (Douglas Peninsula Formation DP, Pearson Formation PA, Et-then Group ET) form an ordered sequence, and the pole for the Epworth basin (Takiyuak Formation TK) falls in the place expected from stratigraphic correlations. Of the poles for the Aphebian rocks of the Labrador Trough only one (CC) might be derived from original magnetizations, and it falls between the post-Huronian tholeiites and the older poles from the Coronation Geosyncline. Other poles from the Labrador Trough fall to the south (RE, SW). Poles from the Late Aphebian rocks of the Richmond Gulf (MRG, RG) and the oldest magnetization at Flin-Flon (FB) fall in south-east South America and the South Atlantic. All these poles fall on a simple polar loop – the *Coronation Loop* of McGlynn & Irving (1978). However, the poles for Stark (ST) and Tochatwi (TP) from within the Athapuscow Aulacogen do not – they fall about 50° to west. This discrepancy is now considered.

The Stark and Tochatwi Formations have been sampled from lakeshore sections distributed over about 4 km just north of the village of Snowdrift (Evans & Bingham 1976; Bingham & Evans 1976). The sections are identified as S and T in Fig. 2. We refer to the immediate region of these lakeshore sections in which both the Stark and Tochatwi have been sampled, as block Z. Block Z is the region from which these westerly poles were obtained. The Stark has not been sampled outside block Z. The Tochatwi has been sampled some 15 km to the east of block Z on Tochatwi Island, but there the magnetizations are secondary (Evans & Bingham 1976). Block Z is characterized by tight folds and steep dips, locally overturned, which contrast with the low or moderate dips and open folds typical of the Athapuscow Aulacogen as a whole. The Stark Formation is generally brecciated.

Bingham & Evans (1976) have suggested that one possible explanation of the westerly positions of the Stark and Tochatwi poles is that they have been rotated clockwise about a local vertical axis. They note that the bedding strikes at the Stark and Tochatwi localities in block Z differ systematically from those in adjacent parts of the aulacogen, and they calculate that an anticlockwise tectonic rotation of 60° would bring the strikes into agreement. This rotation would place the Stark and Tochatwi poles at about 18° S, 83° W. Their prediction is in excellent agreement with results reported here. The predicted position is midway between poles we have found for the Takiyuak (TK) and Pearson (PA) Formations (Fig. 11). The Takiyuak Formation is the correlative of the Tochatwi in the Coronation Geosyncline and is autochthonous (Hoffman 1973). The Pearson Formation is the near time equivalent of the Tochatwi. The Pearson Formation has been sampled palaeomagnetically from an autochthonous region about 8 km north of block Z (McGlynn & Irving 1978). We believe this to be good evidence in favour of the rotation hypothesis as originally suggested on structural grounds by Bingham & Evans (1976).

Recently Gough, Rahman & Evans (1977) have argued, on the basis of measurements of susceptibility anisotropy, that what we refer to as block Z has not undergone local rotation. They recognize that an important problem in interpreting their data concerns the degree to which the original depositional fabric has been retained. They make use of two parameters, q and f (Rees & Frederick 1974; Rees 1971). The anisotropy quotient q is a measure of the shape of the susceptibility ellipsoid; if it is low (high) the susceptibility ellipsoid is oblate (prolate). Values of $q > 0.67$ are taken to indicate that the susceptibility fabric is secondary. f is the angle between the axis of minimum susceptibility (K_3) and the bedding pole, and is usually less than 15° when the susceptibility fabric is undeformed. The parameters q and f have been developed for studies of the fabric of deep ocean sediments in which the origin of the susceptibility and the mode of deformation is likely to be very different from that of shallow marine and fluvial red beds of the Athapuscow Aulacogen. The origin of the susceptibility of the latter is unknown, although Gough *et al.* (1977) have offered some speculations.

The basis of the argument of Gough *et al.* (1977) against rotation is that the directions of (primary fabric) maximum susceptibility K_1 observed in the Snowdrift region are similar to those in the Kahochella Group observed from Keith Island some 50 km to the south-west (localities G and ML of Fig. 2); they use Keith Island as a standard. This procedure could be questioned as Keith Island itself is, in part, allochthonous, and the palaeomagnetic poles indicate that rotation of about 15° could have occurred (see above), but this does not vltiate their comparisons because the effect is not large.

Most of the samples (65 per cent) studied by Gough *et al.* (1977) in the Snowdrift area are from two sections. The Tochatwi section (45 specimens) from Tochatwi Island have K_1 directions similar to those in the Kahochella rocks of Keith Island, and the fabric is considered to be depositional from their q and f values. As noted above this Tochatwi section is not located in block Z and the remanence directions are not depositional but are secondary (Evans & Bingham 1976). The second section is from the Stark Formation (47 specimens) of block Z. It has remanence directions that are regarded as primary (Bingham & Evans 1976), and the K_1 axes are about 60° different from those in the Kahochella. However, K_1 is close to the fold axis and q and f are such that Gough *et al.* (1977) argue that the fabric is secondary. It would appear therefore that the results from these two well-sampled sections provide no information regarding the cause of the westerly displace poles of block Z, because in the one (Tochatwi) the remanence is apparently secondary, and in the other (Stark) the susceptibility fabric is apparently secondary.

The remaining 35 per cent of results from the Snowdrift region are distributed amongst

four sections – two from the Stark and two from the Tochatwi, all from block Z. The sampling is too sparse to yield definitive statistics for q and f, but there is no obvious relationship of K_1 to fold axes, and the susceptibility fabric could be depositional. The remanent magnetization is complex, some components being primary, others secondary (Evans & Bingham 1976; Bingham & Evans 1976). In three of the sections K_1 axes differ from those in the Kahochella and therefore could indicate local rotations. In the fourth section K_1 is parallel to that in the Kahochella and could indicate the absence of rotation. The results are equivocal.

Ultimately, evidence on susceptibility anisotropy could provide important tectonic information but the present evidence neither confirms nor refutes the idea of local rotation of block Z. On the other hand it seems to us that the structural complexity of the region of block Z, and the evidence from remanent magnetization, provide good evidence for supposing that the westerly position of the poles for the Stark and Tochatwi Formations were caused by local rotation of block Z, as originally suggested by Bingham & Evans (1976).

Magnetic overprints and regional uplift

The poles for the magnetizations that are considered to be secondary are listed in Table 3 and are plotted in Fig. 12. The Seton *B* magnetization has a pole (SB) that falls in the same general region as those derived by other workers from secondary magnetizations in the Kahochella Group (KS), the Dogrib dykes (DO) and the Pearson Formation (PC), and with those (ML, DB, LG) obtained by Reid (1972) that are also probably secondary. Pole SB also falls in the same general region as poles from the Nonacho Group, and from the Cape Smith volcanics (CS); the former are certainly secondary (negative fold test by McGlynn *et al.* 1974) and the latter could well be (Fujiwara & Schwarz 1975). Finally, there are poles from widespread metamorphic rocks, some of which fall in the same general region around the present-day Caribbean. These rocks have acquired their magnetization during slow cooling after regional metamorphism to amphibolite or granulite grade. Their magnetization age can only be bracketed within a long interval between the time of metamorphism and final uplift, an interval that could exceed 10^8 yr. The assignment of ages to the magnetization of metamorphic rocks is a very uncertain procedure as the following discussion further illustrates.

The first of these 'metamorphic' poles was obtained by Park (1975) from Flin Flon (FA) to which he assigned an age of -1600 to -1700 Ma. Subsequent studies in south-west Greenland have yielded seven more poles from metamorphic rocks. Two of these (one from Kongarnuit dykes from north of the Nagsugsugtoquidian front by Fahrig & Bridgwater (1976), and one from the Arsuk amphibolite dykes by Piper & Stearns (1976)) have errors of 21° and 27° respectively, and do not meet our minimum reliability criteria. The remaining five poles fall into northern and southern groups. The poles from the Kangamuit dykes (Fahrig & Bridgwater 1976) intruded into Archaean terrain south of the Nagsugsugtoquidian front, from Godthaab rocks (Fahrig & Bridgwater 1976) and from the Itivdleq dykes (Beckman & Mitchell 1976) fall in the Caribbean region, and poles for the Sagdlerssuq gneisses and basic bands (Beckman & Mitchell 1976) and the Arsuk Group metavolcanics (Piper & Stearns 1976) fall in high latitudes. The age for the northern group of poles is probably best given by the K–Ar isochron of -1620 ± 50 Ma of Beckman & Mitchell (1976). If this is correct they fall outside the time interval considered here. The ages of the poles in the southern group have been given by Fahrig & Bridgwater (1976) as about -2500 Ma, but Piper & Stearns (1976, fig. 6a) assign them an age of -1600 to -1700 Ma.

According to Beckman & Mitchell (1976), the pole for the Itivdleq dykes that belongs to the southern group has an age of about −1750 Ma, whereas Piper & Stearns (1976) make a comparison with the well-dated −1830 Ma pole for the Dubawnt Group. Such large differences in absolute age assignments are symptomatic of the difficulties in interpreting absolute ages of magnetization in metamorphic rocks, and we do not therefore propose to consider further these Greenland poles at this stage.

More success has, however, attended the attempts to determine relative ages of secondary poles. Park (1975), from detailed af and thermal studies, has found two magnetizations in amphibolite terrain at Flin-Flon (FB and FA, Figs 11 and 12). FB is older, indicating a north-westerly trend in the younger limb of the Coronation Loop. Morgan (1976) has shown from af studies of metamorphic terrain in south-west Greenland (crosses of Fig. 12), a north-westerly trend of poles from Colombia to northern Mexico, and he provides some thermal evidence to indicate that this represents a trend with time.

All this rather indecisive evidence from secondary magnetizations that were almost certainly acquired long after the rocks were formed can be reconciled by constructing a relatively simple polar path beginning in the South Atlantic at about −1750 Ma and trending generally north-westwards reaching the Caribbean about −1700 Ma (Fig. 12). Support for this dating of the north-westerly trending limb of the Coronation loop is provided by the pole (SP) from the Sparrow dykes. Their magnetization is very probably original (reversals and contact tests) and their probable age from A^{39}/A^{40} studies is −1700 Ma (McGlynn *et al.* 1974).

There are now about 20 poles for the age bracket −1500 to −1250 Ma situated in the central Pacific, and it has been usual to connect them with older poles by a westerly trending path as indicated in Fig. 12. There is, however, much uncertainty about this forward connection (Irving, Emslie & Park 1977). If Beckman & Mitchell (1976) are correct in assigning an age of about −1600 Ma to the northerly group of Greenland poles then it would be reasonable to continue the younger limb of the Coronation loop northward rather than westward. Firmly dated poles in the interval −1700 to −1500 Ma are required before this matter can be settled. This is a major hiatus in the Precambrian palaeomagnetic record from Laurentia.

The poles just reviewed have been obtained from magnetizations that are for the most part of secondary origin, and there seems to be little doubt that a widespread remagnetization episode has affected much of the exposed part of Interior Laurentia over distances of 3000 km. This seems to have occurred about −1700 Ma ago. We follow McGlynn *et al.* (1974) and argue that these overprints were caused by uplift and slow cooling after deep burial and, in some instances, metamorphism. The experimental and theoretical studies of Pullaiah *et al.* (1976) show that such overprints could have been acquired during cooling between about 300 and 200°C in 10^6 or 10^7 yr. The fact that K–Ar ages of about −1700 Ma are widespread shows that general cooling at this time did in fact occur.

In central Ontario overprints (e.g. poles NS and AS of Fig. 11, Roy & Lapointe 1976) seem generally to be older than those from northern Canada and could mean that uplift occurred first in the south. If our calibration of the Coronation loop is correct, then the age of the supposed Nipissing overprint (NS) should be about −1820 Ma, that is, about the same as that of the nearby Sudbury structure as determined from U–Pb measurements on zircon (−1844 ± 3, Krogh & Davies 1975). Most workers now agree that the Sudbury structure was caused by a very large meteorite impact. Moreover, the magnetization of the northern and least deformed rim of the Sudbury structure yields a pole (02 N, 085 W) close to but apparently somewhat older than, that for the Nipissing overprint, after correction assuming the limb to have been originally sub-horizontal (Irving & Lapointe 1975), as is consistent

with the impact theory (Dence 1972). It is possible therefore that the Nipissing overprint was a local phenomenon caused by hydrothermal activity following the Sudbury impact.

Still younger overprints occur in the Athapuscow Aulacogen. The pole for the secondary magnetization of the Douglas Peninsula Formation gives a pole far to the west and near to overprints of the Tochatwi (Fig. 11). Both poles are close to that observed for the Mackenzie diabase, and presumably they arise, as Evans & Bingham (1976) have argued, from cooling following local heating by these massive intrusions.

Summary polar path

This discussion shows that the complex patterns of magnetization observed in middle Proterozoic (about −2200 to about −1700 Ma) rocks from Laurentia can be represented by a rather simple polar path – a southerly trending track followed by a loop (the Coronation loop) and then a north-westerly track extending at least as far as Mexico. This track is consistent with the relative and absolute ages of the rocks studied as they are presently known, and with the order in time of the magnetizations as established by stratigraphy, by contact studies and by their resistance to demagnetization, particularly thermal demagnetization. The physical meaning of this path is that Laurentia was near the pole at about −2100 Ma, it moved into lower latitudes, reaching the equator at about −1800 Ma, and then began moving northwards at the same time undergoing slow uplift that may have been completed earlier in southern Laurentia than in the north. The average rate of latitudinal motion was about 3 cm/yr. The change in direction, or 'hairpin' of Fig. 12, corresponds in time (about −1800 Ma) to the culmination of the major orogeny in the Coronation Geosyncline within the accuracy of the present data. The polar path of Figs 11 and 12 is, we believe, the simplest required to explain the observations. Other, more complex paths are possible, but are not required by the data as they are presently known.

Comparison with Africa and Australia

Piper (1976a, b) has argued that most of the continental crust was once grouped into a single vast continent (Proterozoic Pangaea) during much of the Proterozoic, and has cited palaeomagnetic evidence in its support. It will be our contention that the palaeomagnetic evidence, as it is presently known, is insufficient to argue convincingly for or against such a hypothesis. Piper's argument is that all the Proterozoic palaeomagnetic poles from all continents, when plotted on such a reconstruction, can be joined by a single polar path. We suggest that this is not a good argument because it is always possible to draw a path through a set of poles whose age assignments are subject to the uncertainties (about 100 Ma on average) typical of Precambrian studies. An alternative procedure is to obtain APW paths from each continent, and then to compare them. If the paths are correct, and if there was a single supercontinent, then: (a) all the paths should overlie one another and (b) the path from one continent should neither be longer nor less complex than that from the supercontinent as a whole. A further procedure (c) is to select high-quality well-dated palaeomagnetic poles and to determine if they are consistent with particular continental configurations. We shall now consider these three arguments.

The comparison of APW curves ((a) above) that have been individually constructed for each Precambrian Shield is not straightforward, because of the disparity in quality and quantity between the data from Laurentia and that from the rest of the world. For example, the entire Precambrian of Africa has only yielded seven results that have included thermal demagnetization studies, studies that are fundamental to the interpretation of results from

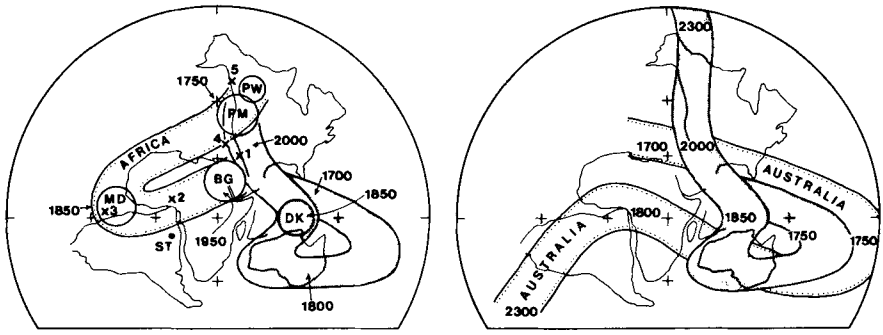


Figure 13. APW paths for about −2300 to −1700 Ma from Australia, Africa and Laurentia compared.

rocks (like most Precambrian rocks) with complex thermal histories. The task becomes even more difficult when the number of observations available are considered. For the time interval −2100 to −1600 Ma there are about 35 well-defined poles from Laurentia that fulfil the minimum reliability criteria, nine from Africa and about six from Australia. Comparisons between such uneven data sets are not easily made, because if differences or similarities are found it is difficult to determine if they are real or arise from inadequacies in the record. Nevertheless, the idea of a Proterozoic Pangaea has gained some currency and palaeomagnetic evidence has been cited in its support (Smith 1976) and it is necessary to present alternative analyses in order to assess the strength of the palaeomagnetic basis for it.

In Fig. 13 the APW paths for Australia (McElhinny & Embleton 1976) and for Africa (Compston & McElhinny 1975) are compared with the path from Laurentia correcting all paths to Piper's Pangaea. These African and Australian polar paths summarize the published information in a reasonable fashion. The African and Laurentian poles agree approximately at about −1900 Ma, but the African path then swings to the west rather than to the south like the Coronation loop; the pole for the Mashonaland dolerites, whose age is estimated to be about −1850 Ma (Compston & McElhinny 1975), is about 90° from several well-established poles at −1850 Ma from Interior Laurentia (Fig. 11). The comparison of the Laurentian and Australian polar path is also made in Fig. 13; they disagree for most of the time interval. Comparison of the two paths of Fig. 13 shows the absence of similarity between the Australian and African curves. In Fig. 13 it is assumed that the polar paths refer to the same (presumed north) pole. If they do not then the comparisons are meaningless. The polarity of all Precambrian polar paths is very uncertain because segments of polar paths cannot be reliability connected, and because the forward connections to the Phanerozoic pole paths cannot yet be unambiguously made. For example, there are major uncertainties in the forward connections of the polar curve for Laurentia at −1700 Ma (see above) at −1200 Ma (Irving *et al.* 1977), and at −600 Ma (Irving & McGlynn 1976b). This factor adds an ambiguity of 180° to the intercontinental comparisons of Precambrian poles! We therefore conclude that either some or all of the APW paths between −2100 and −1700 Ma are wrong, or that Piper's Pangaea did not exist for all this interval. It has already shown that the poles at −2200 Ma from Africa and Interior Laurentia are 90° apart on Piper's reconstruction (McGlynn *et al.* 1975), and that there is no substantial basis for determining from palaeomagnetism the relative positions of continents in the interval −1500 to −1300 Ma because there are only four published pole positions from all Precambrian Shields other than Laurentia for this interval (Irving *et al.* 1977). Hence, in our opinion, there is no basis in the comparison of APW curves constructed independently from

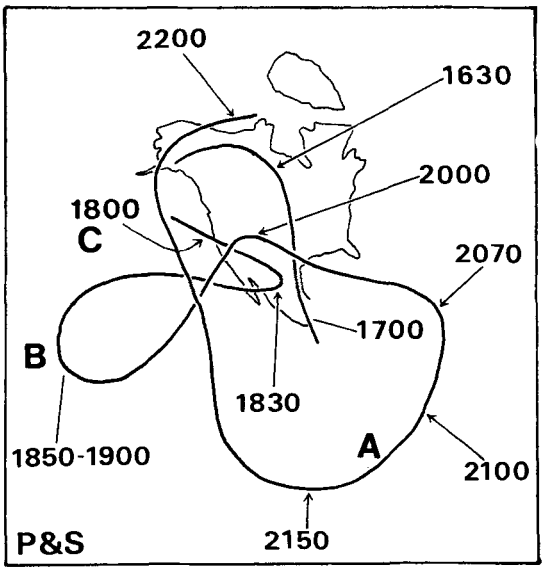


Figure 14. APW path for North America compiled by Piper & Stearns (1976).

different continents for assembling the continents in the interval −2200 to −1300 Ma into a single land-mass in the manner suggested by Piper.

The second line of argument ((b) above) is now considered. Piper & Stearns (1976) and Piper (1976a) have constructed a single APW path for the interval −2200 to −1600 Ma using results from both Africa and Laurentia (Fig. 14). The African results have been rotated into the North American frame according to Piper's supercontinent. The data are mainly North American; for example, in the interval −1850 to −1600 Ma only 18 per cent of the results are from Africa. Their proposed path is in two parts, the first from −2200 to −1800 Ma, and the second from −1700 to about −1600 Ma. The total length is about twice that of our path, and their path has a different form. The differences among Figs 11 to 14 represent, in a nutshell, the palaeomagnetic basis for two competing concepts of global tectonics in the middle Proterozoic. Figs 11, 12 and 13 provide the basis for the notion of continents moving relative to one another more or less continuously in Phanerozoic fashion, whereas Fig. 14 summarizes the argument used by Piper to support his hypothesis of a fixed Pangaea during the whole interval −2200 to −1600 Ma. What are the reasons for these differences in interpretation?

Our path (Figs 11 and 12) and that of Piper & Stearns (Fig. 14) are in approximate agreement at about −2200 Ma and from about −1700 to −1600 Ma. But at other times there are three points of conflict, labelled A, B and C in Fig. 14. Detailed and extensive studies have now established that between −2200 and −1900 Ma the pole for Laurentia lay in western North America between Alaska and the Gulf of Mexico (Fig. 11). No well-dated palaeomagnetic poles for this interval for which there is evidence that the magnetization is primary have yet been found to fall in the southerly loop A of Fig. 14. However, evidence of much younger secondary magnetization, and of magnetization of very uncertain age occur in loop A, and these have been used by Piper (1976a), we believe erroneously, as evidence for loop A. First, there are some poles for the Nipissing diabase (−2150 Ma, Fig. 11, NS) but Roy & Lapointe (1976) have argued on the basis of detailed studies that the magnetization is secondary and that the pole for the magnetization acquired during intrusion of the Nipissing diabase was situated in western North America (Fig. 11, NP). Likewise poles from

subordinate magnetizations observed in Abitibi diabase (−2147 Ma) fall in loop A, but again recent detailed studies of contact rocks have shown that the pole (AP) corresponding to the magnetization acquired at the time of intrusion is in fact near California (Irving & Naldrett 1977), and the subordinate magnetizations could be overprints (Roy & Lapointe 1976). Thirdly, the pole from a preliminary survey of the Indian Harbour dykes have been cited by Piper (1967a) as having a corresponding age of 2080 Ma (K–Ar), in fact, a much more detailed survey of these *same* bodies (= Michael Gabbro) has yielded a different direction and a Rb–Sr isochron of −1488 Ma (Fahrig & Larochelle 1972); these rocks are much younger than −2000 Ma. Finally, the Duck Lake sill also gives a pole in loop A and has been used by Piper as support for it: the Duck Lake sill result is based on five samples only, it does not fulfil minimum reliability criteria, its age is known only within very wide limits (K–Ar ages −1490 to −2090 Ma) and the relationships of its magnetic age to its geological history are entirely unknown. Thus, all the Laurentian data that have been used by Piper (1967a) in support of the southerly loop A of Fig. 14 are based on a preliminary result with poor age control or on magnetizations that could be secondary. Any concrete evidence that there may be for loop A must, therefore, derive from Africa only.

For the time interval −1850 to −1900 Ma, Piper & Stearns (1976) have drawn loop B (Fig. 14). The Laurentian poles from the Stark and Tochatwi Formations (Fig. 11, ST and TP) fall on loop B. In this paper we provide evidence supporting the suggestion of Bingham & Evans (1976) that the Stark and Tochatwi poles have been deflected by local tectonic rotations, but the outcome of this discussion is as yet uncertain. Until this matter is settled these poles cannot be realistically used to construct a polar path.

According to Piper & Stearns the evidence for placing the pole near San Francisco (C in Fig. 14) at −1800 Ma is provided by poles from the Scourie dykes and the Waterberg red beds. Although Beckman (1976) prefers an age of −1850 Ma for the magnetization of the dykes, the evidence he gives is not strong, and ages of about −2000 Ma and approximately −1700 Ma are also possibilities (see above). The age of the Waterberg System is known only within broad limits (Piper & Stearns give −1950 to −1750 Ma). It has yielded five poles spaced over 50°C of arc. The age assignment of −1800 Ma is subject to possible errors of order 10^8 yr. Moreover, the stratigraphically established directions of polar movement for Laurentia after about −1850 Ma are to the south and not the north-west as Piper & Stearns suggest (compare Figs 11 and 14). The evidence from Laurentia indicates that at the −1800 Ma the pole was not near San Francisco but was in fact in southern South America (Fig. 11).

The reason for the apparent conflict between Figs 13 and 14 should now be clear. There is no concrete evidence from Laurentia for loops A and B as drawn by Piper & Stearns, and their pole path at −1800 Ma is insecurely calibrated and inconsistent with results from Laurentia. Numerous very detailed case histories from Laurentia have now shown that multicomponent magnetization is a ubiquitous phenomenon in early Proterozoic terrains (Reid 1972; Park 1975; Evans & Bingham 1976; Bingham & Evans 1976; Roy & Lapointe 1976; Irving & McGlynn 1976; etc.). Moreover, the Stark and Tochatwi poles are probably deflected by local rotations. Once these facts are recognized it becomes possible to reconcile all the published results from Laurentia to a simple polar path, and there is no necessity to construct the complex path of Piper & Stearns (1976).

We now consider the third argument (c). Two pairs of palaeomagnetic poles matched by age have been used to draw the maps of Fig. 15. Their magnetizations have been studied in some detail, and their ages determined to accuracies of 50 or 100 Ma. The poles have been used to determine the latitude and azimuth of Africa and North America. Longitudes are arbitrary. The other continents are added for comparison only. The reconstructions are

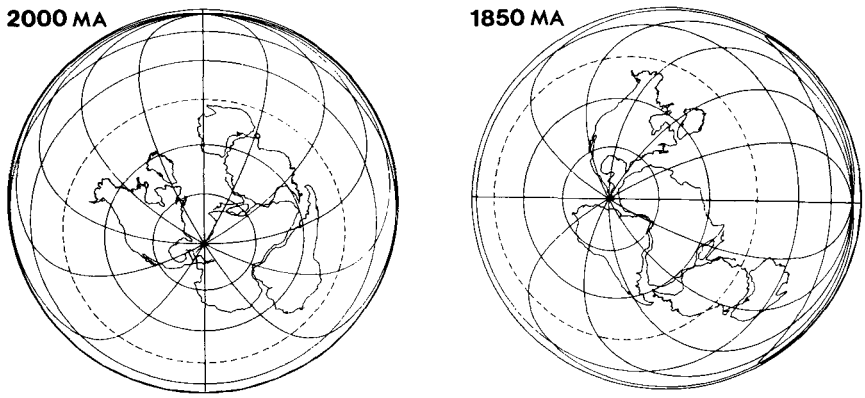


Figure 15. Possible palaeocontinental maps for two time intervals. The poles used for the right map (1850 Ma) are for Africa and Mashonaland dolerite (Compston & McElhinny 1975) and for North America the Dubawnt Group (Park, Irving & Donaldson 1973). The poles for the left map (2000 Ma) are for Africa and Bushveld Gabbro (Gough & Van Niekerk 1959) and for North America the Indin dykes (McGlynn & Irving 1975).

not intended to be realistic, but to show that certain high-quality data are consistent with pangaeas different from that of Piper. It could be argued that other choices of polar pairs would yield different results, but numerous such graphical experiments provide no substantial support for Piper's reassembly (Morel & Irving 1979).

In conclusion, we wish to stress that we do *not* argue that pangaea-type reassemblies of the continental crust did *not* occur during the interval -2200 to -1300 Ma. Our argument is that no substantial case can be made on palaeomagnetic grounds for the existence of a fixed pangaea for the *whole* of this time interval as Piper (1976a, b) has argued. It is possible that, during the Proterozoic, assembly of most of the continent crust may have occurred several times. The lifetime of each pangaea may have been tens or hundreds of millions of years, comparable to the lifetime of Wegener's Pangaea. With continents moving over a globe of finite surface area this is bound to happen sooner or later. Within each successive pangaea the relative positions of the individual cratonic nucleii are likely to have changed. Such ephemeral clusterings of the continental crust would not be detectable with the present density and accuracy of dating of palaeomagnetic poles. Ultimately it should be possible to draw global precambrian palaeocontinental maps and palaeomagnetism will be an important ingredient in such maps. At present, however, our record is inadequate for global reconstructions.

The existing palaeomagnetic record does, however, have some bearing on regional tectonics. It is our opinion that the palaeomagnetic and geological evidence indicate that Laurentia has remained a coherent unit during the interval -2200 to -1300 Ma, that it has undergone considerable internal and marginal deformation, and that it has moved as a whole relative to the pole by very large distances (McGlynn *et al.* 1975; Christie, Davidson & Fahrig 1975; Irving & McGlynn 1976b and therein). There is, however, no palaeomagnetic evidence from this interval of time for extending this idea by combining all the major Precambrian shields into a pangaea for the *whole* of this time. Rather, it seems to us, that the evidence favours the idea that the older continental nucleii have moved about relative to one another more or less continuously during the early Proterozoic, perhaps with the repeated assembly of pangaeas. A possible difference between early Proterozoic and Phanerozoic tectonics was that in the former, considerable *internal* deformation appears to have occurred within the cratons, whereas during the latter deformation was largely

marginal. The preponderance of internal over marginal deformation may, however, be more apparent than real, because nowadays internal deformation is widespread (in much of central and eastern Asia, for example, Molnar & Tapponier 1975) and is more likely to be preserved subsequently, being less vulnerable to later tectonic activity than marginal deformation. It may be a function of chance preservation.

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Earth Physics Branch Contribution No. 753.

References

Baadsgaard, H., Morton, R. D. & Olade, M. A. D., 1973. Rb–Sr isotope age for the Precambrian lavas of the Seton Formation, East Arm of Great Slave Lake, Northwest Territories, *Can. J. Earth Sci.*, **10**, 1579–1582.

Baragar, W. R. A. & Robertson, W. A., 1973. Fault rotation of palaeomagnetic directions in Coppermine River lavas and their revised pole, *Can. J. Earth Sci.*, **10**, 1519–1532.

Beckman, G. E., 1976. A palaeomagnetic study of part of the Lewisian complex, north-west Scotland, *J. Geol. Soc. London*, **132**, 45–69.

Beckman, G. E. & Mitchell, J. G., 1976. Palaeomagnetic and geochronological work in central West Greenland, *Earth Planet. Sci. Lett.*, **30**, 269–280.

Bingham, D. K. & Evans, M. E., 1976. Paleomagnetism of the Great Slave Supergroup, Northwest Territories, Canada: the Stark Formation, *Can. J. Earth Sci.*, **13**, 563–578.

Campbell, F. H. A. & Cecile, M. P., 1975. Report on the geology of the Kilohigok Basin, Goulburn Group, Bathurst Inlet, N.W.T., *Geol. Surv. Can. Paper 75-1A*, pp. 297–306.

Christie, K. W., Davidson, A. & Fahrig, W. F., 1975. The paleomagnetism of Kaminak dikes – no evidence of significant Hudsonian plate motion, *Can. J. Earth Sci.*, **12**, 2048–2065.

Compston, W. & McElhinny, M. W., 1975. The Rb–Sr age of the Mashonaland dolerites of Rhodesia and its significance for paleomagnetic correlation in Southern Africa, *Precamb. Res.*, **2**, 305–315.

Cox, A. & Doell, R. R., 1960. Review of paleomagnetism, *Geol. Soc. Am. Bull.*, **71**, 645–768.

Dence, M. R., 1972. Meteorite impact craters and the structure of the Sudbury Basin, *Geol. Assoc. Can. Spec. Paper*, No. 10, pp. 7–18.

Donaldson, J. A., Irving, E., McGlynn, J. C. & Park, J. K., 1973. Drift of the Canadian Shield, in *Implications of Continental Drift to the Earth Sciences*, pp. 3–18, Volume 1, eds Tarling, D. H. & Runcorn, S. K., Academic Press, London.

Dressler, D., 1975. Lamprophyres of the north-central Labrador Trough, Quebec, Canada, *Neues Jahrb. Min. Mh.*, **6**, 268–280.

Ermanovics, I. & Fahrig, W. F., 1975. The petrochemistry and paleomagnetism of the Molson Dikes, Manitoba, *Can. J. Earth Sci.*, **12**, 1564–1575.

Evans, M. E. & Bingham, D. K., 1973. Paleomagnetism of the Precambrian Martin Formation, Saskatchewan, *Can. J. Earth Sci.*, **10**, 1485–1493.

Evans, M. E. & Bingham, D. K., 1976. Paleomagnetism of the Great Slave Supergroup, Northwest Territories, Canada: the Tochatwi Formation, *Can. J. Earth Sci.*, **13**, 555–562.

Fahrig, W. F. & Bridgwater, D., 1976. Late Archean and Early Proterozoic paleomagnetic pole positions from West Greenland, in *The Early History of the Earth*, pp. 427–439, ed. Windley, B. F., J. Wiley, London.

Fahrig, W. F. & Chown, E. H., 1973. The paleomagnetism of the Otish gabbro from north of the Grenville Front, *Can. J. Earth Sci.*, **10**, 1556–1564.

Fahrig, W. F., Irving, E. & Jackson, G. D., 1971. Paleomagnetism of the Franklin diabase, *Can. J. Earth Sci.*, **8**, 455–467.

Fahrig, W. F. & Jones, D. L., 1969. Paleomagnetic evidence for the extent of Mackenzie igneous events, *Can. J. Earth Sci.*, **6**, 679–688.

Fahrig, W. F. & Larochelle, A., 1972. Paleomagnetism of the Michael Gabbro, and possible evidence of the rotation of the Makkovik subprovince, *Can. J. Earth Sci.*, **9**, 1287–1296.

Fisher, R. A., 1953. Dispersion on a sphere, *Proc. R. Soc. London A*, **217**, 295–305.

Fraser, J. A., 1974. The Epworth Group Rocknest Lake area, District of Mackenzie, *Geol. Surv. Can., Paper 73–39*, pp. 1–23.

Fraser, J. A., Hoffman, P. F., Irvine, T. N. & Mursky, G., 1972. The Bear Province, in *Variation in Tectonic Styles in Canada*, pp. 453–504, eds Price, R. A. & Douglas, J. W., Geol. Assoc. of Can., Special Paper No. 11.

Fujiwara, V. & Schwarz, E. J., 1975. Paleomagnetism of the Circum-Ungava Fold-belt: Cape Smith komatiitic basalts, *Can. J. Earth Sci.*, **12**, 1785–1793.

Gough, D. I. & van Niekerk, C. B., 1959. A study of the paleomagnetism of Bushveld gabbro, *Phil. Mag.*, **4**, 126–136.

Gough, D. I., Rahman, Aziz-ur & Evans, M. E., 1977. Magnetic anisotropy and fabric of red beds of the Great Slave Supergroup of Canada, *Geophys. J. R. astr. Soc.*, **50**, 685–697.

Hoffman, P. F., 1968. Stratigraphy of the Lower Proterozoic (Aphebian), Great Slave Supergroup, East Arm of Great Slave Lake, District of Mackenzie, *Geol. Surv. Can., Paper 68–42*, pp. 1–93.

Hoffman, P. F., 1973. Evolution of an early Proterozoic continental margin of the Coronation Geosyncline and associated aulacogens of the northwestern Canadian Shield, *Phil. Trans. R. Soc. Lond. A*, **273**, 547–581.

Hoffman, P. F., Bell, I. R., Hildebrand, R. S. & Thorstad, L., 1977. Geology of the Athapuscow Aulacogen, east arm of Great Slave Lake, District of Mackenzie, in Report of Activities, Part A, *Geol. Surv. Can., Paper 77-1A*, pp. 117–129.

Irving, E., Donaldson, J. A. & Park, J. K., 1972. Paleomagnetism of the Western Channel diabase and associated rocks, Northwest Territories, *Can. J. Earth Sci.*, **11**, 30–42.

Irving, E., Emslie, R. F. & Park, J. K., 1977. Paleomagnetism of the Harp Lake Complex and associated rocks, *Can. J. Earth Sci.*, **14**, 1187–1201.

Irving, E. & Lapointe, P. L., 1975. Paleomagnetism of Precambrian rocks of Laurentia, *Geosc. Can.*, **2**, 90–98.

Irving, E. & McGlynn, J. C., 1976a. Polyphase magnetization of the Big Spruce Complex, Northwest Territories, *Can. J. Earth Sci.*, **13**, 476–488.

Irving, E. & McGlynn, J. C., 1976b. Proterozoic magnetostratigraphy and the tectonic evolution of Laurentia, *Phil. Trans. R. Soc. Lond. A*, **280**, 433–468.

Irving, E. & Naldrett, A. J., 1977. Paleomagnetism in Abitibi Greenstone Belt and in Abitibi and Marachewan diabase dikes, *J. Geol.*, **85**, 157–176.

Irving, E., Park, J. K. & McGlynn, J. C., 1972. Paleomagnetism of the Et-Then Group and Mackenzie diabase in the Great Slave Lake area, *Can. J. Earth Sci.*, **9**, 744–755.

Krogh, T. E. & Davies, G. L., 1975. The age of the Sudbury nickel irruptive, *Carnegie Inst. Wash. yr Book*, **73**, 567–569.

Larochelle, A., 1966. Paleomagnetism of the Abitibi dike swarm, *Can. J. Earth Sci.*, **3**, 671–683.

McElhinny, M. W. & Embleton, B. J. J., 1976. Precambrian and early Paleozoic palaeomagnetism in Australia, *Phil. Trans. R. Soc. Lond. A*, **280**, 417–431.

McGlynn, J. C., Hanson, G. N., Irving, E. & Park, J. K., 1974. Paleomagnetism and age of Nonacho Group sandstones and associated Sparrow dikes, District of Mackenzie, *Can. J. Earth Sci.*, **11**, 30–42.

McGlynn, J. C. & Irving, E., 1975. Paleomagnetism of early Aphebian dykes from the Slave Structural Province, Canada, *Tectonophys.*, **26**, 23–28.

McGlynn, J. C. & Irving, E., 1978. Paleomagnetism of the Pearson Formation, Great Slave Supergroup, N.W.T. and the Coronation loop, *Can. J. Earth Sci.*, **15**, 642–654.

McGlynn, J. C., Irving, E., Bell, K. & Pullaiah, G., 1975. Paleomagnetic poles and a Proterozoic supercontinent, *Nature*, **255**, 318–319.

McMurry, E. W., Reid, A. B. & Evans, M. E., 1973. A paleomagnetic study of the Kahochella Group, N.W.T., Canada, *EOS Trans. Am. geophys. Un.*, **54**, 248.

Molnar, P. & Tapponnier, P., 1975. Cenozoic tectonics of Asia: effects of a continental collision, *Science*, **189**, 419–426.

Morel, P. & Irving, E., 1979. Tentative paleocontinental reconstructions for the Early Paleozoic and Proterozoic, *J. Geol.*, **86**, 535.

Morgan, G. E., 1976. Palaeomagnetism of a slowly cooled plutonic terrain in western Greenland, *Nature*, **259**, 382–385.

Park, J. K., 1973. Paleomagnetism of metamorphic rocks of Daly Bay and Melville Peninsula, Churchill Structural Province, *Can. J. Earth Sci.*, **10**, 1079–1088.

Park, J. K., 1974. Paleomagnetism of miscellaneous Franklin and Mackenzie diabase of the Canadian Shield, and their adjacent country rocks, *Can. J. Earth Sci.*, **11**, 1012–1019.

Park, J. K., 1975. Paleomagnetism of the Flin Flon–Snow Lake greenstone belt, Manitoba and Saskatchewan, *Can. J. Earth Sci.*, **12**, 1272–1291.

Park, J. K., 1977. A reconnaissance paleomagnetic survey of the central Labrador Trough, Quebec, *Can. J. Earth Sci.*, in press.

Park, J. K., Irving, E. & Donaldson, J. A., 1973. Paleomagnetism of the Precambrian Dubawnt Group, *Geol. Soc. Am. Bull.*, **84**, 859–870.

Piper, J. D. A., 1976a. Palaeomagnetic evidence for a Proterozoic supercontinent, *Phil. Trans. R. Soc. Lond. A*, **280**, 469–490.

Piper, J. D. A., 1976b. Definition of pre-2000 m.y. apparent polar movements, *Earth planet. Sci. Lett.*, **28**, 470–478.

Piper, J. D. A. & Stearns, J. E. F., 1976. Paleomagnetism of Ketilidian metamorphic rocks of SW Greenland, and 1850–1600 m.y. apparent polar movements, *Phys. Earth planet. Int.*, **13**, 143–156.

Pullaiah, G., Irving, E., Buchan, K. L. & Dunlop, D. J., 1976. Magnetization changes caused by burial and uplift, *Earth planet. Sci. Lett.*, **28**, 133–143.

Rees, A. I., 1971. The magnetic anisotropy of samples from the Deep Sea Drilling Project Leg I, *Mar. Geol.*, **11**, M16–M23.

Rees, A. I. & Frederick, D., 1974. The magnetic fabric of samples from the Deep Sea Drilling Project Legs I–VI, *J. Sedim. Petr.*, **44**, 655–662.

Reid, A. B., 1972. Paleomagnetic study at 1800 million years in Canada, unpublished *PhD thesis*, University of Alberta, Calgary, Alberta.

Robertson, W. A., 1964. Palaeomagnetic results from northern Canada suggesting a tropical Proterozoic climate, *Nature*, **204**, 66–67.

Robertson, W. A., 1969. Magnetization directions in the Muskox intrusion and associated dikes and lavas, *Geol. Surv. Can. Bull.*, **167**, 1–51.

Robertson, W. A. & Baragar, W. R. A., 1972. The petrology and palaeomagnetism of the Coronation sills, *Can. J. Earth Sci.*, **9**, 123–140.

Roy, J. L. & Lapointe, P. L., 1976. The paleomagnetism of Huronian red beds and Nipissing diabase; Post-Huronian events and apparent polar path for the interval – 2300 to – 1500 Ma for Laurentia, *Can. J. Earth Sci.*, **13**, 749–773.

Roy, J. L., Reynolds, J. & Sanders, E., 1973. An alternating field demagnetizer for rock magnetism studies, *Publs. Earth Phys. Br. Dep. Energy, Mines Res., Ottawa*, **44**, 37–45.

Roy, J. L., Robertson, W. A. & Keeping, C., 1969. Magnetic "free field" spaces for palaeomagnetism, rock magnetism and other studies, *Can. J. Earth Sci.*, **6**, 1312–1316.

Roy, J. L., Sanders, E. & Reynolds, J., 1971. Un four electrique pour des proprietes magnetiques des roches, *Publs. Earth. Phys. Br. Dep. Energy, Mines Res., Ottawa*, **42**, 229–237.

Schwarz, E. J., 1976. Paleomagnetism of the Circum-Ungava Belt: east coast of Hudson Bay, in Report of Activities, Part B, *Geol. Surv. Can., Paper 76-1B*, pp. 37–38.

Schwarz, E. J., 1977. Paleomagnetism of Proterozoic rocks of the Circum-Ungava Fold Belt, *IAGA/IAMAP*, Seattle.

Smith, P. J., 1976. Does the key fit the locks of the past? *Nature*, **260**, 746–747.

Stewart, A. D. & Irving, E., 1974. Palaeomagnetism of Precambrian sedimentary rocks from NW Scotland and the apparent polar wandering path of Laurentia, *Geophys. J. R. astr. Soc.*, **37**, 51–72.

Symons, D. T. A., 1966. A paleomagnetic study of the Gunflint, Mesabi and Cuyana iron ranges in the Lake Superior region, *Econ. Geol.*, **61**, 1336–1361.